

Ömer Sayilgan

Control Systems & GNC Engineer

Eschenbachstraße 9
Munich
Germany

+49 155 10742256

✉ omersayilgan56@gmail.com



Profile

Control and GNC engineer with hands-on experience across the full loop: controller design in MATLAB/Simulink, code generation to embedded targets, and commissioning on real hardware. Combines classical control with industrial automation — **TwinCAT/Beckhoff PLC** work alongside **HIL** validation and Python data infrastructure. M.Sc. Aerospace at TU Munich, **completing December 2026**; **available from December 2026**, based in Munich.

Professional Experience

- 08/2025–
03/2026 **Working Student — Mechatronics & Automation, Phlair, Munich, Germany**
- Developed **TwinCAT 3.1 (Structured Text)** control logic for an absorber test bench, from I/O mapping through commissioning.
 - Integrated and wired **RS-485** industrial sensors to **Beckhoff I/O terminals** from circuit diagrams, and calibrated them over serial protocols.
 - Built an automated **PLC-to-InfluxDB** data pipeline in **Python (PyADS)**, replacing manual measurement logging.
 - Designed real-time **Grafana/InfluxQL** dashboards used by the team to monitor test-bench runs live.
 - Deployed the supporting database and monitoring stack on **Docker, Proxmox** and Linux networking.
- 08/2023–
05/2024 **Control Systems Engineer, Roketsan Inc., Ankara, Turkey**
- Designed **PID and robust controllers** for a high-precision **missile seeker gimbal**, covering pointing and line-of-sight stabilisation across the operating envelope.
 - Built and validated **dynamic models** in **MATLAB/Simulink**, tuning controllers against measured plant response.
 - Generated **Embedded C** from Simulink models via **Embedded Coder** and deployed onto **Xilinx** hardware targets.
 - Ran **real-time and Hardware-in-the-Loop (HIL)** test campaigns on **xPC Target** rigs to qualify controller performance before integration.

Engineering Projects

- 2025–2026 **Master's Thesis — Fault-Tolerant Control of Propulsive Spacecraft, TU Munich**
- Built a **modular Python GNC framework** (~23,000 lines) for landing trajectory optimisation and actuator-fault analysis of an Apollo-class lunar lander.
 - Architected it as a **shared core engine** — vehicle and actuator models, optimal-control problem formulation, attainable-moment envelopes — reused by nine self-contained study packages.
 - Ran automated **fault-injection and fault-onset campaigns** across 16 fault modes and multiple initial-condition regimes, auto-generating result tables and figures.

- 2024–2025 **Semester Thesis — Automated Signal-Integrity Check Toolbox**, *Institute of Flight System Dynamics, TU Munich*
- Extended an automated verification toolbox by developing **MATLAB OOP** classes and scripted **Simulink model generation**.
 - Implemented **V-Model-based testing** with code-compliance checking (Mrails) and reusable **test-harness** structures.

Education

- 10/2023–12/2026 **M.Sc. Aerospace**, *Technical University of Munich (TUM)*, Munich, Germany, *expected December 2026*
- **Provisional Grade:** 2.3 — **Credits:** 92 ECTS
 - **Control & GNC:** Advanced Control (2.0), Nonlinear Control (2.0), Optimal Control (2.3), Model Reference Adaptive Control (3.0), System Identification (2.3).
 - **Software & Systems:** Advanced Programming / C++ (**1.7**), Real-Time Systems (2.0), Introduction to ROS (**1.0**), MATLAB/Simulink for CAE (2.0).
 - **Labs:** Flight Control System Lab (**1.7**), Cyber-Physical Systems Lab — Autonomous Applications (1.9).
- 2017–2023 **B.Sc. Mechanical Engineering**, *Middle East Technical University (METU)*, Ankara, Turkey
- Final Grade: 3.33 / 4.0 (German equivalent: **1.7 / “Gut”**) — top 10–15% of the department.
 - **Minor:** Mechatronics (Robotics, Computer Science, Logic Design). Multiple “High Honor” placements.

Technical Skills

Control & GNC	PID, Robust, Nonlinear, Adaptive (MRAC) and Optimal Control (LQR, MPC); Lyapunov methods; Fault-Tolerant Control; System Identification; Trajectory Optimisation; 6-DOF Flight Dynamics
Industrial Automation	TwinCAT 3.1 (Structured Text), Beckhoff PLC, EtherCAT/RS-485 & serial protocols, sensor integration and calibration, test-bench commissioning
Modelling & Test	MATLAB, Simulink, Embedded Coder code generation, Model-Based Design, SIL/HIL testing, xPC Target, V-Model test harnesses
Programming	Python (NumPy, SciPy, PyADS), C, Embedded C, C++, MATLAB OOP
Tools & Infrastructure	Git, Docker, Linux (Ubuntu), Proxmox, InfluxDB, Grafana, ROS (ROS 2 familiar)

Languages

English	Full professional proficiency
German	B1
Turkish	Native

university instruction in English
Goethe-Institut certified, actively improving